

Lab 16: Bayesian Belief Network

Aim

To implement a **Bayesian Belief Network (BBN)** and calculate the probability of an event using conditional probabilities.

Algorithm

1. Define the prior probability of the parent event.
2. Define the conditional probabilities of the child events.
3. Calculate the probability of the observed evidence.
4. Apply Bayes Theorem to calculate the posterior probability.
5. Display the calculated probabilities.

Program

```
# Lab 16: Bayesian Belief Network
```

```
# Prior probability
```

```
P_Rain = 0.3
```

```
# Conditional probabilities
```

```
P_WetGrass_given_Rain = 0.9
```

```
P_WetGrass_given_NoRain = 0.2
```

```
# Probability of no rain
```

```
P_NoRain = 1 - P_Rain
```

```
# Total probability of wet grass
```

```
P_WetGrass = (
```

```
    P_WetGrass_given_Rain * P_Rain
```

```
+ P_WetGrass_given_NoRain * P_NoRain  
)
```

```
# Posterior probability using Bayes Theorem
```

```
P_Rain_given_WetGrass = (  
    P_WetGrass_given_Rain * P_Rain  
) / P_WetGrass
```

```
print("P(Wet Grass) =", round(P_WetGrass, 3))
```

```
print("P(Rain | Wet Grass) =", round(P_Rain_given_WetGrass, 3))
```

Sample Output

P(Wet Grass) = 0.41

P(Rain | Wet Grass) = 0.659

Result

Thus, the **Bayesian Belief Network was successfully implemented**, and the posterior probability of rain given that the grass is wet was calculated.